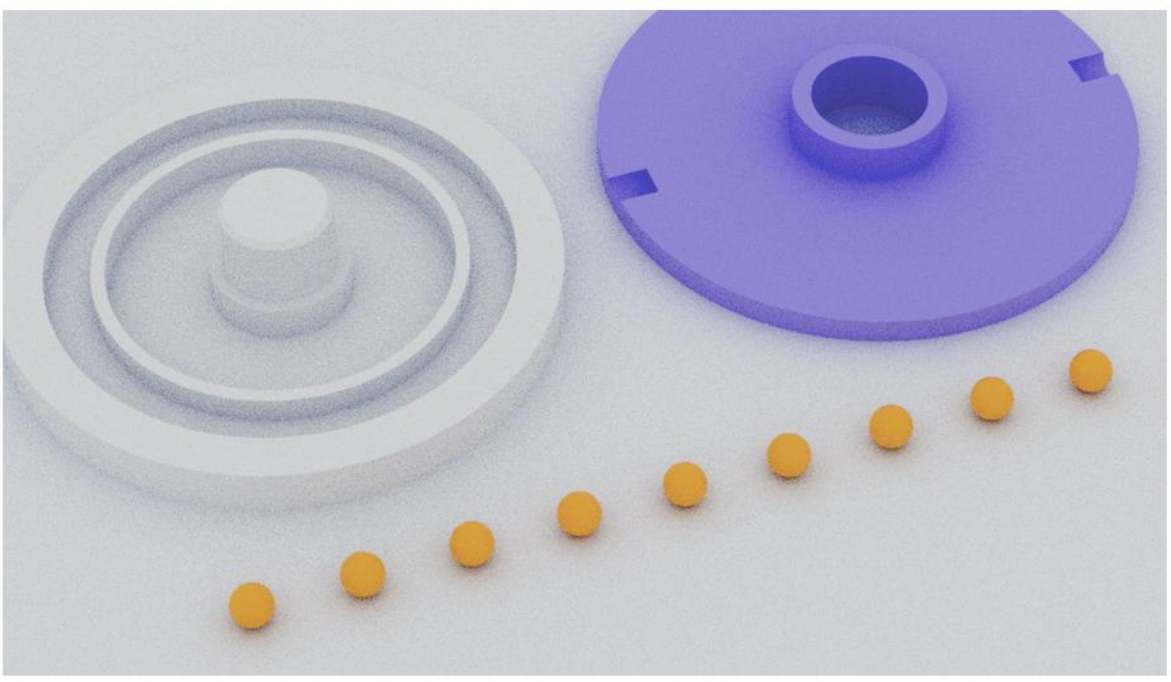
Pie Caddy: bearing test

The caddy's real ball track at 80 mm, and 27 airsoft BBs (6 mm, any brand).

1 What you need

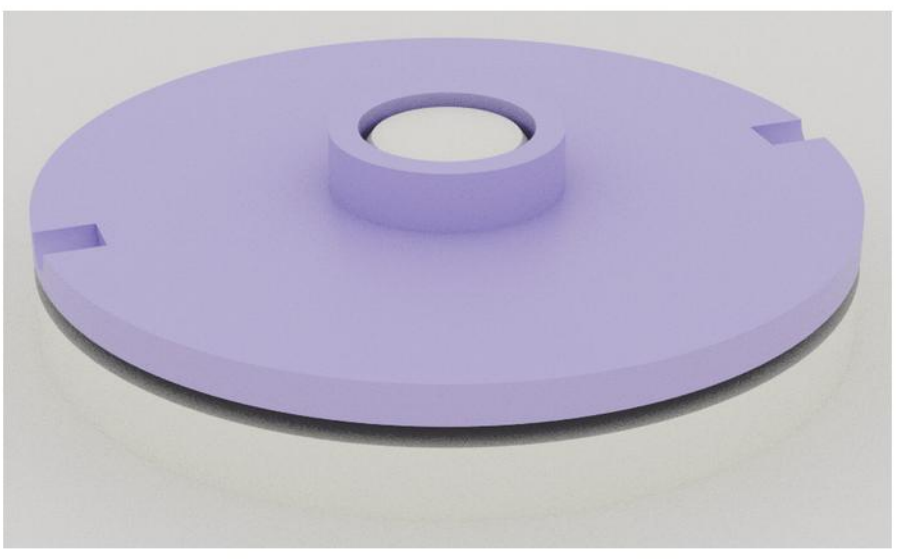


- **The base (white here): a ball track round a centre post.**
- **The top (purple): a disc with a hole for the post.**
- **27 airsoft BBs, 6 mm. Not in the print: a bag of thousands is a few dollars.**

2 Drop the BBs into the track

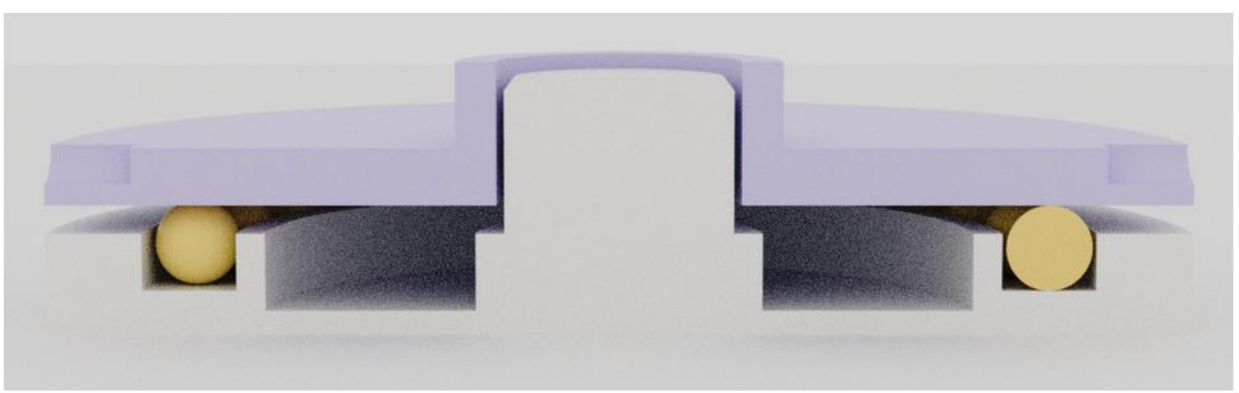
- All 27, all the way round. A small gap is left on purpose so they can't jam.

3 Set the top on



- Its hole over the post, flat side down. It rests on the balls, not on the base.
- Lift it straight off any time.

4 Inside



- The balls roll between the track's floor and the top's smooth underside.

Design-review renders of the real parts, not photos. Colours are only to tell the parts apart.

Pie Caddy: test it

A mug or a few pens on top makes it a fair test.

1 Spin it

- **Flick it: does it spin and coast?**
- Listen: quiet, or does it rumble?
- Watch the edge as it turns: does it wobble?

Good

- Spins with a flick and coasts.
- Quiet.
- No wobble.

Tell me

- if it grinds or rumbles;
- if it stops quickly;
- if it wobbles, or the balls jam.

Design-review renders of the real parts, not photos. Colours are only to tell the parts apart.